

Intelligent Platform for Crowdsourcing

Georgia Institute of Technology (GT)

Vertically Integrated Projects (VIP)

Fall 2023



Introduction

What are we doing here?

- IP Crowd
- Creating crowdsourcing-based tools that factor in human-computer interactions

How are we doing this?

- Working on improving social media through specific software features
- Tackling the issues of addictiveness, argumentativeness, and divisiveness
- Take advantage of the good by reducing the bad

This is a PNAMBC presentation.

Pay No Attention to the Man Behind the Curtain

Discussion Challenge Mode

Members:

- Asim Baral, David Liu, Harshith Gutha, Krish Ranglani, Kun-Lin Hsieh, Maanas Kumar, Max Xu, Suhas Mummigatti, Vanshika Varadh

Discussion Challenge Mode

Problem Statement

- Rise in echo chambers
 - **What is it?** An environment where a user is only exposed to opinions that coincide with their own
 - Limits diverse perspectives and comprehensive discussions
 - Creates polarization that reduces collaboration due to an “us vs them” mentality

Goal

- To reduce the "echo chamber" effect by providing insights from an opposing perspective

Discussion Challenge Mode

Solution:

- A challenge generator that produces counterarguments to the comment the user is about to post and forces the user to acknowledge the counterarguments
 - Counter Argument generated by ChatGPT API call through prompt engineering

Discussion Challenge Mode

Airplane seats are designed to recline and should be used in a reclined position.

Sitting behind a reclined airline seat isn't as big of a deal as the complainers make it out to be. Additionally, any room you 'lose' by the seat in front of you reclining can be made up by just reclining your seat and being a bit more comfortable yourself.

Comments

1: This is a lot more heated of a topic than I expected. Good post OP, something that actually fits the ethos of the sub. I'm pro recline all the way, I don't fly that much (3-5 times a year most years) but my flights are typically at least 5 hours long (I do one Asia trip a year which is usually upwards of 14 hours) and couldn't imagine being upright the whole way.

2: Reclining is taboo? I do it every time. Had no idea. Still going to do it bc I paid for the seat but huh. Never knew

3: FYI, multi-millions go into designing these seats so that when passengers recline, it doesn't encroach on leg space, negatively impact the TV in the seats or your tray table. Recline as much as you want. Doesn't change the spatial reasoning of your seat nearly as much as your emotions are making you think it does. Frequent flyer.

4: If you are over 5'8 its a problem. Reclining does not give you more leg room. And some seats don't recline. The issue is that people are getting bigger and airline seats are getting smaller. Regardless of how you feel about this fact, it remains a fact.

5: I like to watch a movie on my tablet set on the tray table in front of me while I'm on the plane. The person in front of me reclining, and also me reclining, will make it so that I can't do that.

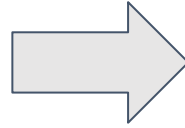
Add Comment

I like to recline and don't think that it affects the person behind me that much.

GPT-3 Response:

Generate Challenge

Submit



Add Comment

I like to recline and don't think that it affects the person behind me that much.

GPT-3 Response:

Is it fair to recline your seat if it prevents the person behind you from using their tray table or watching a movie on their tablet, especially considering the issue of limited legroom in modern airline seats?

Generate Challenge

Submit



4: If you are over 5'8 its a problem. Reclining does not give you more leg room. And some seats don't recline. The issue is that people are getting bigger and airline seats are getting smaller. Regardless of how you feel about this fact, it remains a fact.

5: I like to watch a movie on my tablet set on the tray table in front of me while I'm on the plane. The person in front of me reclining, and also me reclining, will make it so that I can't do that.

6: I like to recline and don't think that it affects the person behind me that much. If meals are being handed out to people, however, I make sure to put my seat back up so that the person behind me can eat their food on their tray table.

Demo

Discussion Challenge Mode

Future Plans:

- How do we make sure the user actually interacts with the counter arguments generated?
 - Currently, there's only an acknowledgment feature, users are not being forced to “resolve” counterarguments.
 - Do we want to enforce this type of control on users?
(overbearing and may push users off the platform)

Opinion Breakdown

- **Members:** Max Xu, Eric Zhang, Sylvia Li, Yoon Ji Cho, Krish Ranglani, Maanas Kumar, Vanshika Varadh, Kun-Lin Hsieh

Opinion Breakdown

Problem Statement: It can be difficult to identify dissenting opinions in discussion-based forums, leading to suppression of differing viewpoints and critical thinking



Benjammin172 • 3 mo. ago

The plant has been operational for quite some time, they're just adding new units to the existing plant and the third just came online. The expansion itself was hilariously over budget, but I doubt too many people will complain if the net result is a smaller bill.



39



Reply



Share



dontfeedthedinosaurs • 3 mo. ago

The bills are getting bigger to fund the debt payments. I like nuclear, but the cost overruns have been ludicrous.



14



Reply



Share



6 more replies



ismelladoobie • 3 mo. ago

Good. Creates permanent jobs and provides clean energy for the whole state. One of the few reasons to be proud to live here.



63



Reply



Share



2 more replies



thecamino • 3 mo. ago

The power plant is good. The apparent mismanagement of funds so far is concerning.



17



Reply

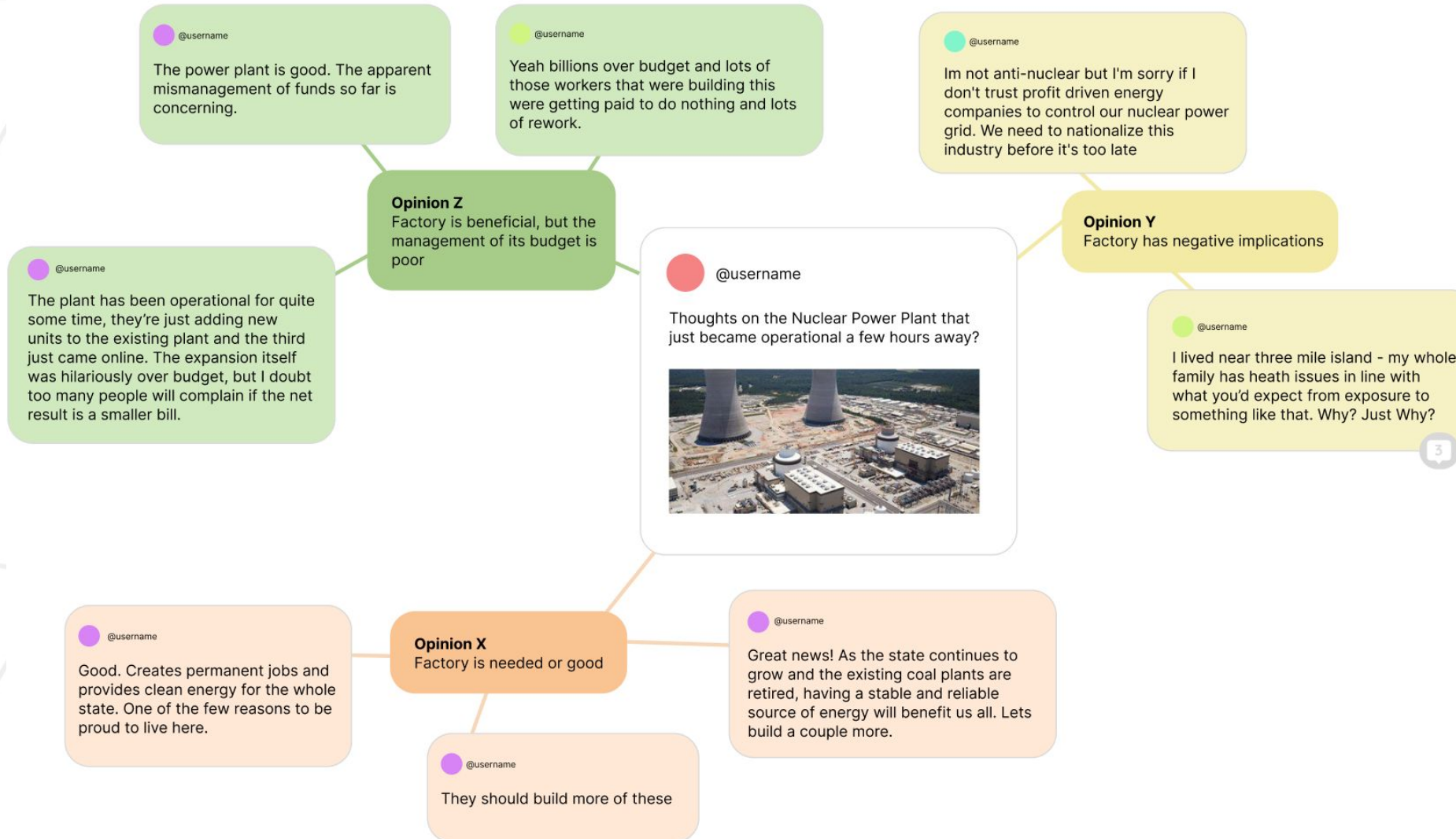


Share



2 more replies

Opinion Breakdown



Opinion Breakdown

Used cosine similarity to compare phrases.

1. Used the HuggingFace tokenizer to extract important words in each comment.
2. Vectorised the phrase using TF:IDF

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

tf_{ij} = number of occurrences of i in j

df_i = number of documents containing i

N = total number of documents

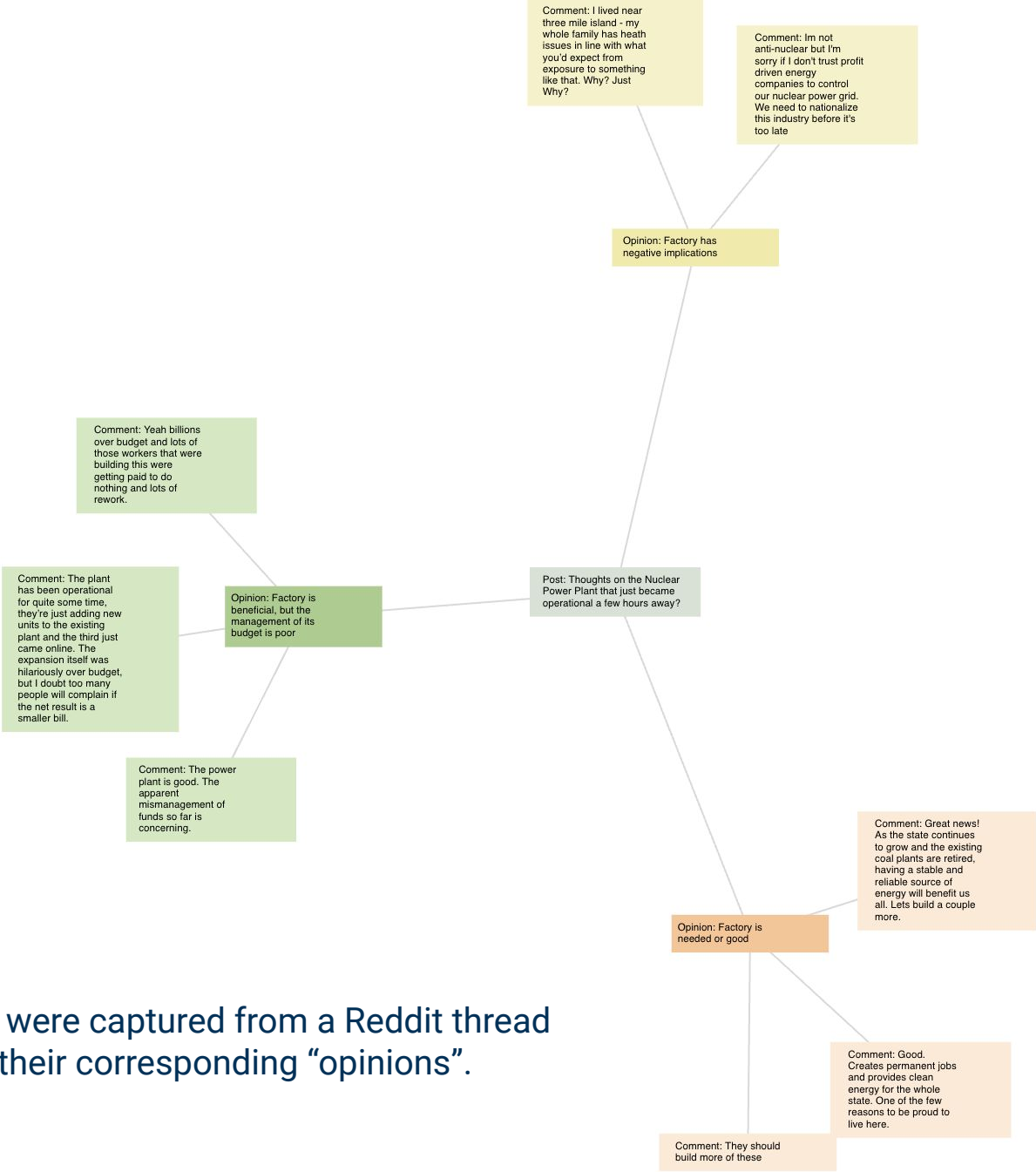
3. Clustered comments using k means.

Opinion Breakdown

1023	Egglordl	nta thank you for protecting your daughter her privacy and l	1	1.63E+09	['nta', 'thank', 'protect', 'daughter', 'privaci', 'belong', 'parent', 'would', 'act', 'like', 'wife', 'would', 'last', 'issu', 'daughter', 'fact', 'good', 'role', 'model', 'show',
1024	Kurt_blow	nta you sound like a damn good father too	1	1.63E+09	['nta', 'sound', 'like', 'damn', 'good', 'father']
1025	Heather_B	the only thing you are ta for is not discussing an appropriate	1	1.63E+09	['thing', 'ta', 'discuss', 'appropri', 'measur', 'wife', 'suppos', 'team', 'work', 'togeth', 'protect', 'daughter', 'convers', '3', 'decid', 'cousin', 'need', 'learn', 'ask', 'p
1026	WeWanna	nta sont pas ben dans tetes tu protege ta fille de la bonne n	1	1.63E+09	['nta', 'sont', 'pa', 'ben', 'dan', 'tete', 'tu', 'proteg', 'ta', 'fill', 'de', 'la', 'bonn', 'manier', 'et', 'tai', 'le', 'seul', 'de', 'son', 'bord', 'ca', 'met', 'en', 'criss', 'de', 'voir', 'c
1027	meowpitb	nta sounds like sammy is a professional mooch himself also	1	1.63E+09	['nta', 'sound', 'like', 'sammi', 'profession', 'mooch', 'also', 'person', 'teenag', 'girl', 'normal', 'behavior', 'especi', 'without', 'ask', 'first', 'huge', 'disrespect', 'da
1028	Angel-407	tell them all straight they need to apologise to you your dau	1	1.63E+09	['tell', 'straight', 'need', 'apologis', 'daughter', 'stop', 'sulk', 'leav', 'within', '7', 'day', 'choic']
1029	python_nc	all you really need to say is my house my rules do not like it	1	1.63E+09	['realli', 'need', 'say', 'hous', 'rule', 'like', 'exit']
1030	pfresh331	1000 nta you are being a good dad and putting her needs fir	1	1.63E+09	['1000', 'nta', 'good', 'dad', 'put', 'need', 'first']
1031		at least now we know why the family did not want the guy a	1	1.63E+09	['least', 'know', 'famili', 'want', 'guy', 'hous']
1032	MeadowA	nta if you did not have zoeys back then no one would i was	1	1.63E+09	['nta', 'zoey', 'back', 'one', 'would', 'kid', 'could', 'keep', 'valuabl', 'home', 'keep', 'stuff', 'absolut', 'way', 'comfort', 'home', 'either']

Opinion Breakdown

The graph view enables easy visualization and interactivity. You can drag the nodes around with a mouse.



These comments were captured from a Reddit thread and grouped into their corresponding “opinions”.

Opinion Breakdown

A single opinion and its supporting comments.

Comment: I lived near three mile island - my whole family has health issues in line with what you'd expect from exposure to something like that. Why? Just Why?

Comment: I'm not anti-nuclear but I'm sorry if I don't trust profit driven energy companies to control our nuclear power grid. We need to nationalize this industry before it's too late

Opinion: Factory has negative implications

Post: Thoughts on the Nuclear Power Plant that just became operational a few hours away?

Demo

Opinion Breakdown

Future Plans

- Create API to connect backend to frontend graph component
- Develop functionality to include sub-comments/additional hierarchy of opinions
- Create normal comment section that links to the graph visualizer

Post-Sentiment Analysis

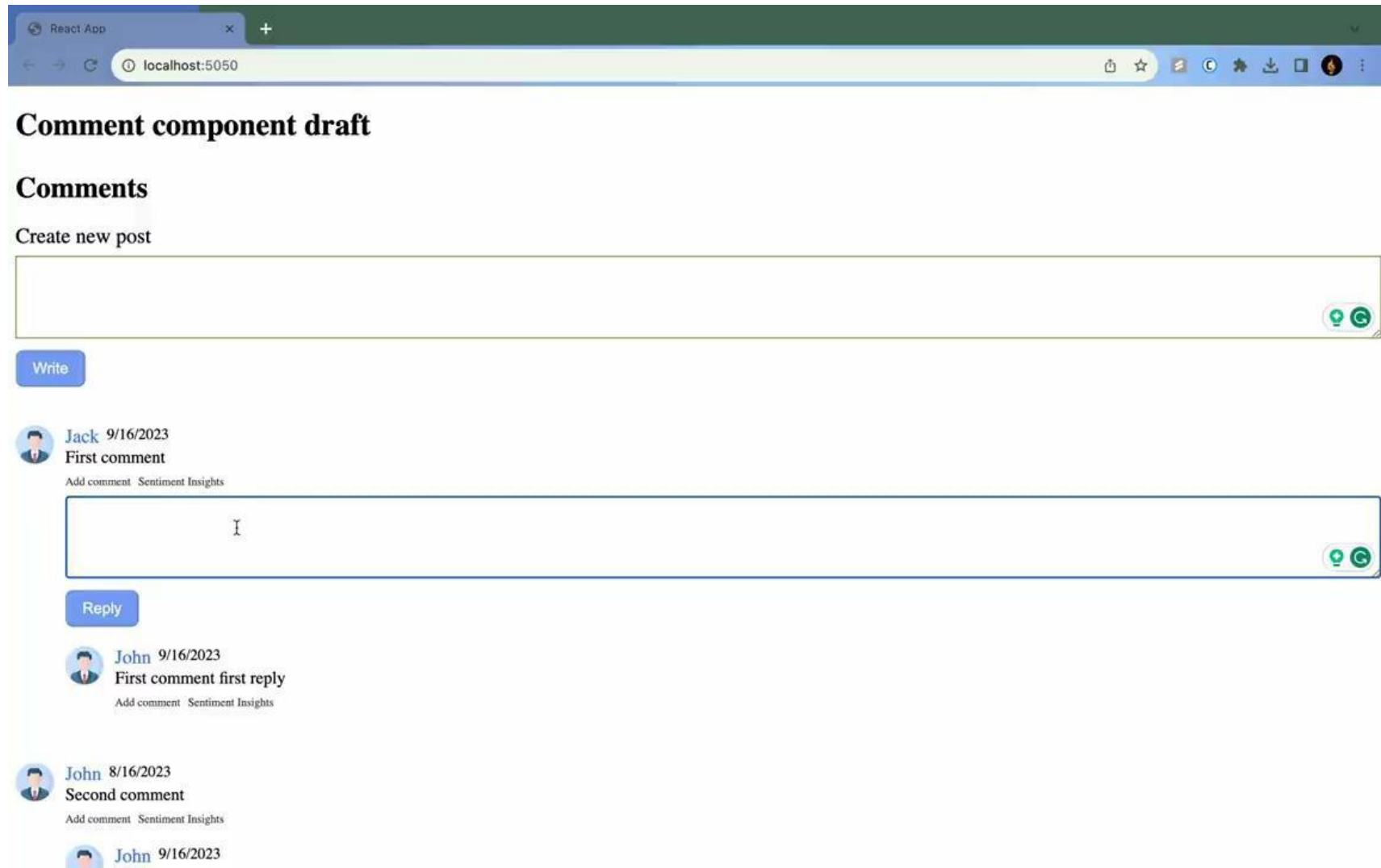
- **Members:** Amith Shetty, Bryan Zeng, Cole Kaplan, David Liu, Harshith Gutha, Max Xu
- **Problem Statement:** Encourage high-value comments through emotion self-monitoring
 - Mediating the tone of a thread/topic
 - Emotional self-awareness to reduce emotionally invested comments that spark heated, non-productive arguments

Post-Sentiment Analysis

- **Feature:** Pop-up for each comment that gives an emotion distribution breakdown
 - Sentiment analysis by RoBERTa transformers ML model
 - Frontend with React and Javascript for comment thread

Post-Sentiment Analysis

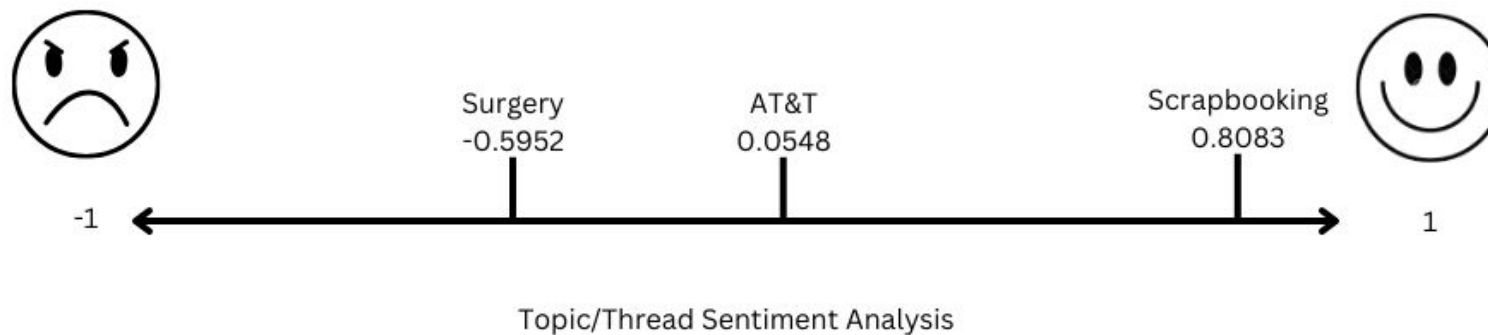
DEMO:



Post-Sentiment Analysis

- **Future Plans**

- Different visualizations of emotion distribution such as pie chart, etc.
- User-independent emotional habit breakdown
- Emotion distribution for entire thread



Thumbs-Up

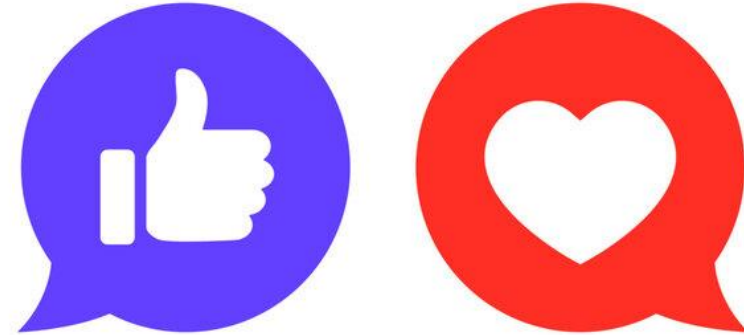
Members:

Sarah Chow,
Willa Zheng,
Chris Lin,
Asim Baral,
Suhas Mummigatti

Thumbs-Up

Members: Sarah Chow, Willa Zheng, Chris Lin, Asim Baral, Suhas Mummigatti

Problem Statement: Thumbs-Up reactions do not properly convey the varied emotions an user might feel when reading a post.



Thumbs-Up

Methods: We conducted a survey, asking audiences about some of their own interaction techniques on social media, normally, and feedback on the mockup of our techniques.

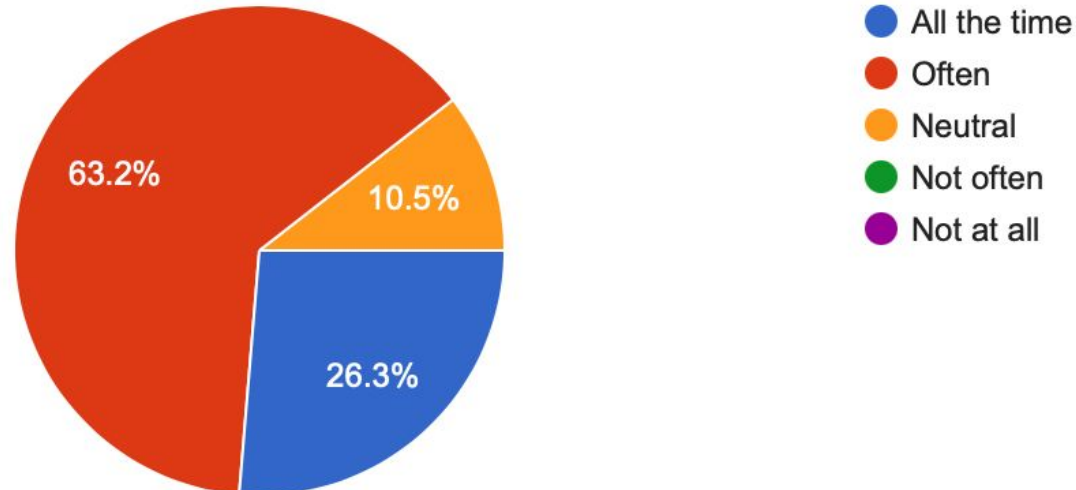


Thumbs-Up

Results:

When reading online content, do you often find yourself agreeing with some parts while disagreeing with others?

19 responses



About 73.7% of the survey respondents indicated that often or all the time, they do not agree with all parts of a post on social media.

Thumbs-Up

Our Solution: A more customizable interaction technique which allows users to choose which part of a post they want to interact with, as well as the opportunity to react with a more nuanced approach. Our new interaction technique must be easily adaptable and provide users a chance to respond in a detailed manner.

Thumbs-Up

Features:

1. Annotation feature with multiple highlight colors corresponding to different reactions
2. Aggregated highlights from the reader's perspective with attention drawn to busy sections of the post
3. Filtering between highlights of different reactions: like, disagree, amusing, important, informative, etc

leads to the privatization of water resources for marginalized communities. Overall, the convenience of water bottles comes at a significant cost to the environment and society, making them a problematic choice in the context of sustainability and conservation.

Like

Disagree

Important

Informative

Amusing

Water bottles are often criticized for their negative environmental impact. One of the primary reasons they are considered bad is their contribution to plastic pollution. The majority of water bottles are made from non-biodegradable plastic, which takes hundreds of years to decompose, filling landfills and oceans with harmful waste. Additionally, the production of plastic bottles consumes vast amounts of fossil fuels and releases greenhouse gases, contributing to climate change. Moreover, the extraction and transportation of bottled water require substantial energy resources, further exacerbating environmental issues. This practice also leads to the privatization of water resources, making clean water less accessible to marginalized communities. Overall, the convenience of water bottles comes at a significant cost to the environment and society, making them a problematic choice in the context of sustainability and conservation.



Important

Thumbs-Up

Water Bottles

Select text to highlight and add reactions:

Water bottles are often criticized for their negative environmental impact. One of the primary reasons they are considered bad is their contribution to plastic pollution. The majority of water bottles are made from non-biodegradable plastic, which takes hundreds of years to decompose, filling landfills and oceans with harmful waste. Additionally, the production of plastic bottles consumes vast amounts of fossil fuels and releases greenhouse gases, contributing to climate change. Moreover, the extraction and transportation of bottled water require substantial energy resources, further exacerbating environmental issues. This practice also leads to the privatization of water resources, making clean water less accessible to marginalized communities. Overall, the convenience of water bottles comes at a significant cost to the environment and society, making them a problematic choice in the context of sustainability and conservation.

Agree

Disagree

Interesting

Funny

Thumbs-Up

View reactions of other people

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Thumbs-Up

[Demo](#)

Thumbs-Up

Future Plans

- Experiment with different categories of reactions and find which ones people are more likely to use
- Determine how to scale the highlight gradient as reactions increase and annotations become cluttered
- Quoting feature for users' annotations to be quoted into their comments to provide additional context for their reaction
- Making the color highlights more accessible and offering different color palettes or design for color-blind users
- Improvement of visual appeal and overall UX through user testing

Future Challenges and Directions

- Problems of interest (define your problem)
 - Addiction
 - Detachment
 - Information Reduction/Overload
 - Transparency
- Discuss future challenges of social media; what are we trying to address and potentially solve?
- Groups formed (e.g. addiction, transparency, etc.)
- Potential strategies and features to implement on each group

Detachment

Members: Max Xu, Kun-Lin Hsieh, Amith Shetty

Description: Users are overwhelmed with an ocean of posts that are pivotal events, which may lead to users becoming desensitised to important events.

Problem: People elect influential political figures, so it's in society's best interest for people to actively care about what's happening in the world.

Potential Solutions:

1. Show a popup when users scroll too long on a certain popup and ask users to type what they learned
2. Force users to take break from the app if using it for too long
3. Show fewer articles from topics that the user has read a lot from
4. Show distracting articles that are unrelated to the topic that the user has seen.

Information Reduction

Members: Cole Kaplan, Harshith Gutha, Sarah Chow, Suhas Mummigatti, Bryan Zeng

Problem:

When encountering vast amounts of information, it may be detrimental to the user as it may not be relevant or potentially be overwhelming. As such, it would be beneficial to find ways to reduce this amount of information to better represent the user's interests.

Proposed Solutions:

- Use of more filters, to more accurately find information
- Remove previously viewed posts from feed
- Promote trustworthy sources, credibility system
- Archival of old or outdated posts
- Limits in place, such as time limit or post limit

Addiction

Members: Yoon Ji Cho, Willa Zheng, Vanshika Varadh, Asim Baral, Eric Zhang

Problem:

When using social media, many users grow addicted to it to the point where it starts to affect their day to day life.

Heavy screen time and exposure to social media can lead to increased risk of mental illnesses and other undesirable effects on mental health.

This is especially true for children and young adults under the age of 21 that have brains that are still developing.

Proposed Solutions:

- Use of a recommendation engine that will recommend “boring posts”
 - Less visually appealing
 - Less attuned to their interests
- Removing interaction features
- More buffers and ads as people use the platform

Transparency

Problems:

1. Is it possible to remain anonymous online while getting more refined recommendations?
2. The users do not know how their data is being used.
3. The users do not know what data is being used and collected.

Proposed Solutions:

1. Highlight the important points in the user agreement relating to data use.
2. Allow the user to change the data that was collected to give more accurate recommendations and also see which parameters are collected.
3. After each recommendation use display based on which factors the recommendation was given.

Questions?